IoT-Enabled Real-Time Patient Monitoring and Alert System for Intraoperative and Postoperative Care

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I. Introduction

A. Introductory Statement

Continuous measurement of vital signs such as heart rate, peripheral oxygen saturation (SpO_2) and body temperature is a routine part of patient care during surgery (the intraoperative phase) and during recovery (the postoperative phase). Deviations from safe ranges must be noticed quickly by medical staff, who are often responsible for several patients at once and are not always at the bedside.

In parallel, the Internet of Things (IoT) has become a key enabling technology for healthcare. The Internet of Medical Things (IoMT) is described in recent volumes on remote healthcare systems and applications [1], IoT- and ICT-based architectures for patient care [2], biomedical sensors and signal processing [3], the secure exchange of real-time health data [4] and wearable and connected devices [5], and in reviews of enabling technologies and architectures [6] and of biosensors integrated with multi-hop IoT systems through cloud connectivity [7]. Much of this material is presented as concepts, architectures and application overviews. This paper proposes an IoT-enabled real-time patient monitoring and alert system and reports its problem statement, research questions, objectives and methodology. The system is implemented as a working web-based prototype in which the IoT sensor nodes are emulated by virtual sensors, so that the complete monitoring workflow can be demonstrated and tested without hardware.

B. Background of the Study (Literature Review)

IoMT systems and applications. Hemanth et al. [1] collect work on remote healthcare systems and applications of IoMT, and the volume on IoT and ICT for healthcare [2] presents applications and architectures that improve health services. A recent volume on the medical IoT [5] focuses on wearable technologies and connected devices and discusses both their opportunities and the technological advances still needed. Aledhari et al. [6] review enabling technologies, architectural elements, challenges and future directions of biomedical IoT, and Uddin and Koo [7] review biosensors integrated with multi-hop IoT systems via cloud connectivity for real-time remote patient monitoring. These works show that remote monitoring is a central IoMT use case, but they are surveys rather than a single reproducible, tested implementation.

Sensing and security. The volume on IoT for healthcare technologies [3] covers medical imaging and signal processing, biomedical sensors and biotechnological advances; failure detection and alert design are outside its main focus. Blockchain-based approaches to secure IoMT data are discussed in [4].

Networking foundations. Kurose and Ross [8] give the theory of delay, loss, timeouts and application-layer protocols, which underlies the failure-detection reasoning of this work. The MQTT specification [9] defines a publish/subscribe protocol with quality-of-service levels and a Last Will mechanism that signals abnormal disconnection; it is the intended transport once physical devices are added.

AI in healthcare. Books on artificial intelligence in healthcare [10], [11] cover data, ethics, remote monitoring and the security and privacy of health data. They position machine-learning analytics as a possible extension, whereas the alerting in this work is rule-based and transparent.

Prototype studies. Naeem et al. [12] present an ESP32-based system for hospitalized high-risk COVID-19 patients in which vital signs are sent through a gateway to an IoT server and SMS alerts are issued when readings pass predetermined thresholds. Krishnapriya and Prema [13] build a low-cost system with an ESP32 board and a MAX30102 sensor that sends heart rate and SpO_2 to a cloud IoT platform; a pilot with volunteers reports an average capture-to-display latency of 1–2 s, and the authors note that Wi-Fi links can be unreliable. These studies confirm that dashboards and threshold alerts are an accepted pattern, but behaviour under interruptions is discussed rather than tested systematically. Table I summarizes the gaps and how this work differs.

C. Problem Statements

The core problem is that the pipeline carrying vital signs from the patient to medical staff is not yet unified, fault-

TABLE I
COMPARISON OF RELATED SOURCES AND THIS WORK

Source	Limitation / gap	How this work differs
IoMT volumes [1], [2], [5]	Overview level; no testable end-to-end implementation	One working, testable monitoring workflow
Sensing volume [3]	Failure detection and alerting are not the main focus	Adds watchdog and threshold-alert logic
Blockchain for IoMT [4]	Security-centred; overhead for time-critical alerts	Lightweight prototype; security as future work
Reviews [6], [7]	Surveys; no implementation or tests of their own	Implements part of the reviewed architecture
Naeem et al. [12]	Gateway algorithm and SMS alerts; no outage-handling tests reported in the abstract	Missing-data watchdog, per-patient thresholds, automated tests
Krishnapriya and Prema [13]	HR and SpO_2 only; pilot-level validation; Wi-Fi reliability noted as a limit	Adds temperature, multi-patient dashboard, history, alert log, watchdog
Kurose and Ross [8]; MQTT [9]	General-purpose theory and protocol	Timeout-based failure detection applied to patient monitoring
AI books [10], [11]	AI-oriented; no monitoring-software design	Rule-based alerts; AI as future work

aware and self-explanatory. Three research problems (RP) are addressed:

- **RP1:** Continuous vital-sign data may not be available to medical staff in one remote interface. IoT sensors and a central dashboard are needed for near-real-time monitoring.
- **RP2:** Network delays or interruptions can reduce the timeliness of patient monitoring data. The system should transmit data continuously and detect delayed or missing updates.
- **RP3:** Large data streams may be difficult to interpret quickly without automated alerts. A dashboard should visualize data and alert staff when values cross set thresholds.

Anaesthetists and surgical and recovery-ward nurses are directly affected, and ultimately patients, whose safety depends on early detection of abnormal readings.

D. Research Questions

- **RQ1:** How can an IoT system collect and display patient vital signs in near real time?
- **RQ2:** To what extent can reliable data handling maintain timely monitoring during interruptions?
- **RQ3:** How can visualization and threshold alerts improve identification of abnormal readings?

E. Research Objectives

1) General Objective: To design, develop, implement, and evaluate an IoT system for real-time remote patient monitoring during intraoperative and postoperative care, including data transmission and alerts. In the current prototype the sensor nodes are emulated by virtual sensors.

2) Specific Objectives:

- **Objective 1 (RQ1):** To design and implement a web application with virtual sensor nodes that collects, stores and displays heart rate, SpO_2 and temperature of several patients on one dashboard, showing each submitted reading within 100 ms and updating simulated readings automatically at a configurable interval (2 min by default).
- **Objective 2 (RQ2):** To develop and test a watchdog and persistence mechanism that detects within 7 s (5 s timeout plus a 2 s check interval) that a device has stopped sending, rejects readings from offline devices with a logged warning, restores the stream on reconnection, and keeps all stored data after a page reload.
- **Objective 3 (RQ3):** To implement configurable per-patient thresholds with trend charts and automatic alerts (banner, log, sound, acknowledgement) raised within 100 ms of an out-of-range reading and re-evaluated when a threshold changes.

Each problem maps to one question and one objective (RP1→RQ1→Objective 1, and so on), and each objective is verified by automated tests of the prototype.

II. METHODOLOGY

A. Software Development Methodology

To ensure a systematic, rigorous and controlled approach to building the system, an **Agile, iterative development process** (a lightweight Scrum adapted to a single developer) is used throughout the semester, as shown in Fig. 1. Each iteration follows the cycle Plan, Design, Build, Test and Review and ends with a working version of the application. The methodology was chosen because the requirements for alert rules, dashboard views and stored data were refined after each version of the prototype, and because short iterations always leave a runnable application that can be tested. In the research the phases are used as follows: *Plan*: select the next backlog items from the MVP features and link them to a research question; *Design*: define screens, the patient data model and the alert and watchdog rules; *Build*: implement the single-page web application (HTML, CSS, JavaScript, charts on an HTML canvas); *Test*: run automated browser tests (Chromium, Playwright) that record measurements for the evaluation; *Review*: compare the results with the acceptance criteria derived from the objectives.

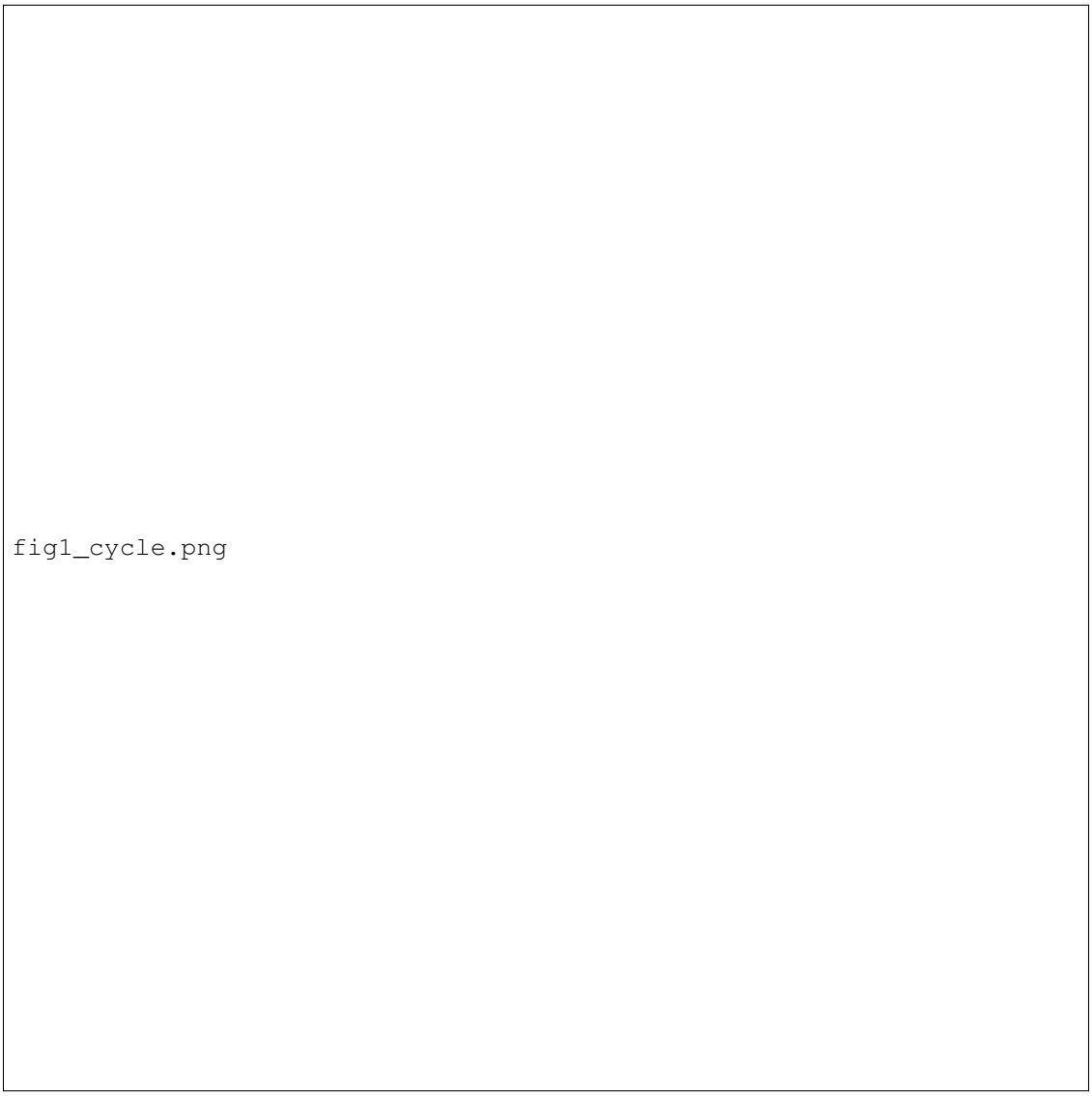


Fig. 1. Circular Agile cycle (Plan, Design, Build, Test, Review) and iteration history and plan.

B. System Architecture

The system follows a **three-layer client-side web architecture** with a pluggable storage layer (Fig. 2). The layers separate user presentation, core processing logic and data persistence; the underlying principles are modularity and separation of concerns.

- **Presentation / input layer:** patient manager (add, edit, delete), manual entry form, device and auto-simulation controls, and the dashboard with patient cards, live trend charts, alert banner and history tab.
- **Business logic / processing layer:** the virtual sensor generates simulated readings (2 min by default; 30 s and 10 s demonstration modes); the reading handler accepts readings only from online devices and keeps the last 500; the threshold and alert engine compares each reading with the patient's limits; the watchdog raises a missing-data alert after 5 s without data, checked every 2 s.
- **Data / storage layer:** a storage adapter saves each patient as one document, in a shared document database when the hosting platform provides one and otherwise in browser localStorage.

Data flow: data originate from the manual entry form or the virtual sensor, pass through the reading handler, the alert engine and the watchdog, are saved by the storage adapter, and end at the dashboard (cards, charts, alerts and history).

C. Data Model

A **document-oriented (NoSQL) model** is used: one document per patient in a collection named *patients* (Fig. 3). Each document holds scalar fields (identifier, name, phase, ward, device state and revision counter) and embedded structures for the current vitals, the thresholds (defaults: heart rate 50–120 bpm, $SpO_2 \geq 92\%$, temperature 35.5–38 °C), up to

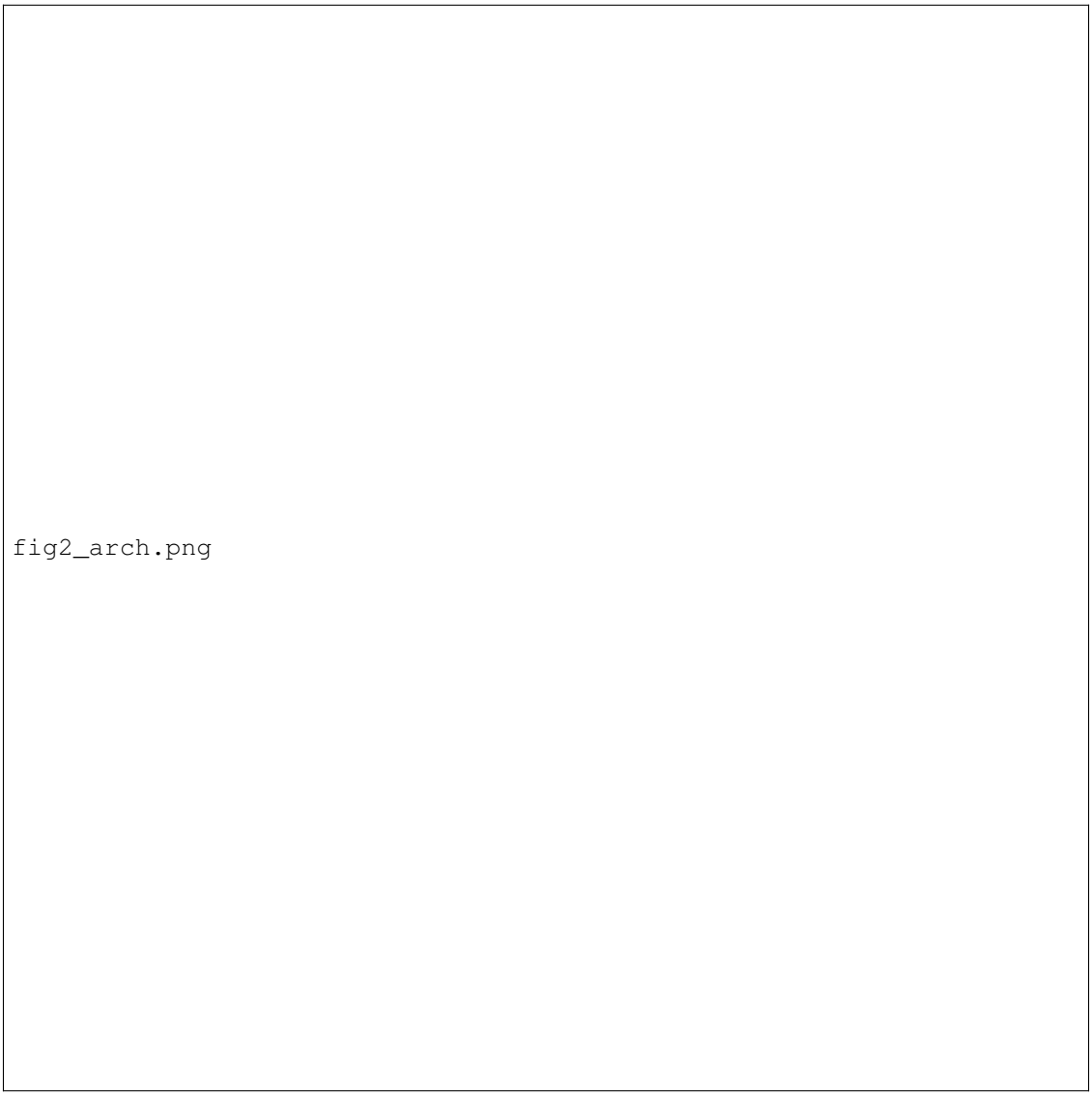


Fig. 2. Three-layer architecture of the web-based monitoring prototype.

500 readings, up to 30 alerts and flags of active alerts. The model was selected because each patient's data are small and always read together, a document can be written atomically, and no server or schema migration is needed, which keeps the prototype self-contained.

D. MVP Scope

The Minimum Viable Product (MVP) is focused exclusively on the core problem statement. **In scope (implemented):** F1 patient management; F2 manual entry and virtual-sensor auto-simulation; F3 dashboard with patient cards, trend charts with threshold lines and a live clock; F4 per-patient thresholds and alerts with banner, log, sound and acknowledgement; F5 device control and a watchdog; F6 history of stored readings; F7 persistent storage and automated functional tests. **Out of scope (future work):** physical ESP32 sensor nodes and MQTT transport, a server-side backend with a relational or time-series database, user authentication, machine-learning prediction, hospital information-system integration and notifications by SMS or e-mail.

E. Evaluation Plan

The prototype is evaluated with automated functional tests that run in a Chromium browser (Playwright) against the application in localStorage mode. Alert display times are measured with the browser's high-resolution timer; the watchdog and auto-update tests use a virtual clock. Table II lists the metrics and the targets derived from the objectives.

F. Project Timeline

Table III gives the roadmap, aligned with the iterations in Fig. 1; completed items are marked as such and the rest are planned.

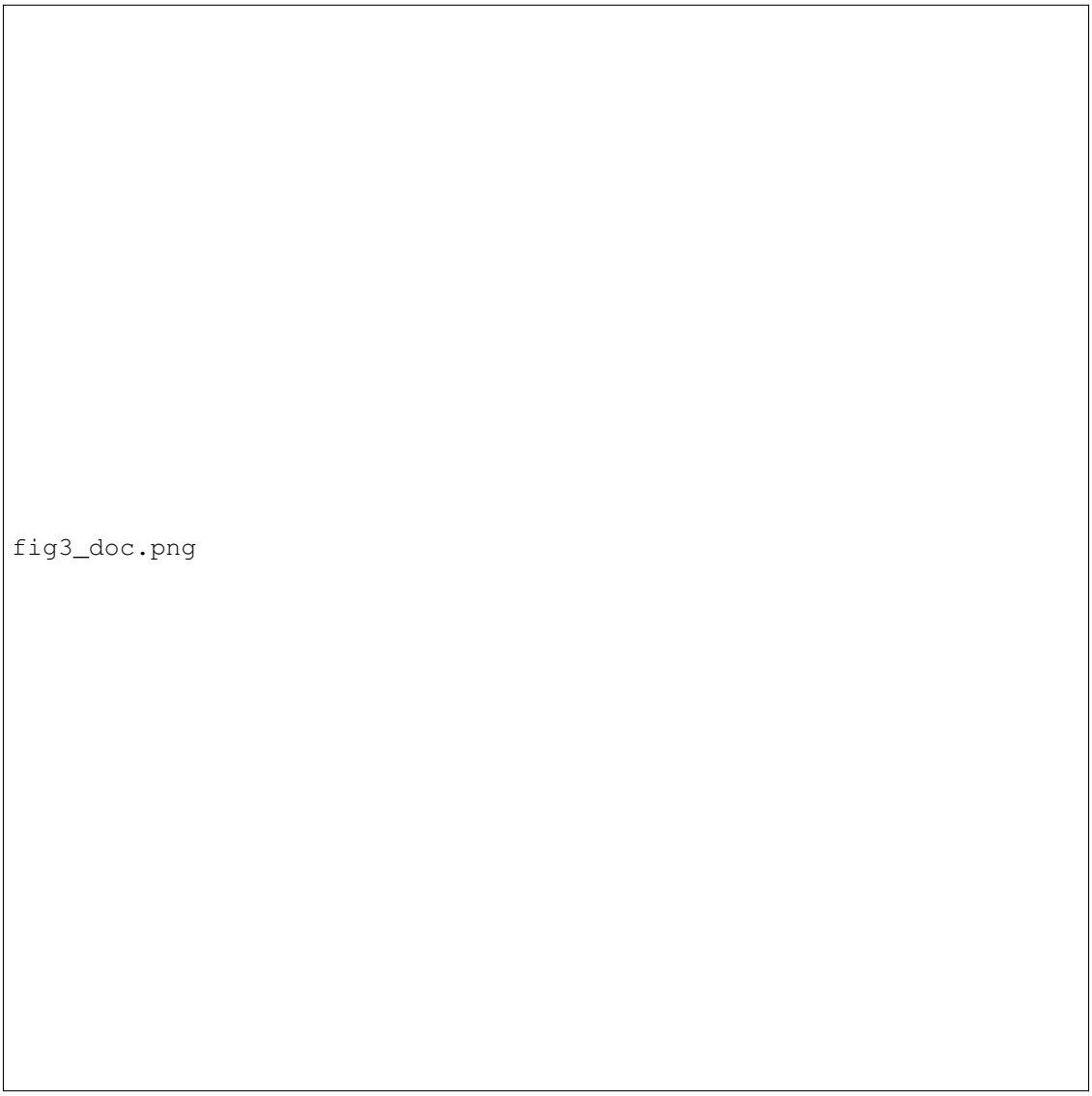


Fig. 3. Document model of the patients collection.

G. Ethics, Security and Limitations

No real patient data are used: patients are entered as pseudonymous codes and all readings are manual or simulated. The prototype has no user authentication, and data in browser storage stay on the device; authentication and encrypted transport are required before any real use. The system is a research prototype and must not be used for clinical decisions. Its main limitations are the emulated sensors, the single-user storage in browser mode and the absence of clinical validation.

REFERENCES

[1] D. J. Hemanth, J. Anitha, and G. A. Tsihrintzis, Eds., *Internet of Medical Things: Remote Healthcare Systems and Applications*. Cham, Switzerland: Springer, 2021, doi: 10.1007/978-3-030-63937-2.

[2] *IoT and ICT for Healthcare Applications* (EAI/Springer Innovations in Communication and Computing). Cham, Switzerland: Springer, 2020, doi: 10.1007/978-3-030-42934-8.

[3] C. Chakraborty, A. Banerjee, M. H. Kolekar, *et al.*, Eds., *Internet of Things for Healthcare Technologies* (Studies in Big Data, vol. 73). Singapore: Springer, 2021, doi: 10.1007/978-981-15-4112-4.

[4] *Blockchain for Secure Healthcare Using Internet of Medical Things (IoMT)*. Cham, Switzerland: Springer, 2023, doi: 10.1007/978-3-031-18896-1.

[5] *Advancing Healthcare with the Medical Internet of Things: Revolutionizing Patient Care through Connected Devices* (Health Informatics). Cham, Switzerland: Springer, 2026, doi: 10.1007/978-3-032-23933-4.

[6] M. Aledhari, R. Razzak, B. Qolomany, A. Al-Fuqaha, and F. Saeed, "Biomedical IoT: Enabling technologies, architectural elements, challenges, and future directions," *IEEE Access*, vol. 10, pp. 31306–31339, 2022, doi: 10.1109/ACCESS.2022.3159235.

[7] R. Uddin and I. Koo, "Real-time remote patient monitoring: A review of biosensors integrated with multi-hop IoT systems via cloud connectivity," *Appl. Sci.*, vol. 14, no. 5, Art. no. 1876, 2024, doi: 10.3390/app14051876.

[8] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 8th ed., Global ed. Harlow, U.K.: Pearson Education, 2022.

[9] OASIS, "MQTT Version 5.0," OASIS Standard, Mar. 2020. [Online]. Available: https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.html

[10] P. Mahajan, *Artificial Intelligence in Healthcare: AI, Machine Learning,*

TABLE II
EVALUATION METRICS AND TARGETS

Metric	**Method**	**Target**
Alert display time (RQ3)	Out-of-range manual readings; time from submission to alert shown	$\leq$ 100 ms
Threshold re-evaluation (RQ3)	Lower a limit below the current value	Alert raised
Acknowledgement (RQ3)	Acknowledge an alert in the log	Marked
Missing-data detection (RQ2)	Devices disconnected at different moments	$\leq$ 7 s
Stream restoration (RQ2)	Reconnect a device	Logged
Persistence (RQ2)	Reload the page	No data lost
Auto-update interval (RQ1)	Virtual sensor for several minutes of virtual time	120 s

TABLE III
IMPLEMENTATION ROADMAP

Weeks	**Deliverable / milestone**	**Status**
1–3	Problem definition, literature, research questions and objectives	Completed
4–6	First simulation; architecture and data model design	Completed
7–8	Web application with alerts, watchdog, history, persistence; automated tests	Completed
9–12	Extended tests, usability review, bug fixing	Planned
13	Research evaluation: answers to RQ1–RQ3	Planned
14	Final report, demonstration and presentation	Planned

and Deep and Intelligent Medicine Simplified for Everyone, 3rd ed. (Academic Edition). Albuquerque, NM, USA: MedMantra, 2022.

[11] A. Bohr and K. Memarzadeh, Eds., *Artificial Intelligence in Healthcare*. London, U.K.: Academic Press, 2020.

[12] S. Q. Naeem, A. I. Majeed, and N. N. ALfaisaly, "Design and implementation of monitoring system for lethal events of high-risk COVID-19 patients," *Iraqi J. Elect. Electron. Eng.*, vol. 21, no. 1, pp. 221–231, 2025, doi: 10.37917/ijeee.21.1.22.

[13] G. Krishnapriya and S. Prema, "Cost-effective IoT-based real-time vital sign monitoring: An affordable telehealth solution," *Telehealth Med. Today*, vol. 10, Art. no. 611, 2025, doi: 10.30953/thmt.v10.611.